

Exploring Redesigning Digital P2P Payments to Facilitate Social Connections: A Participatory Design Approach

Lingyuan Li
Clemson University
Clemson, USA
lingyuanli9936@gmail.com

Guo Freeman
Clemson University
Clemson, USA
guof@clemson.edu

Wen Duan
Clemson University
Clemson, USA
wend@clemson.edu

ABSTRACT

Existing work has highlighted the increasingly important role of digital peer-to-peer (P2P) payments (i.e., money transactions between individuals via online or mobile services) in our social lives. Indeed, such transactions continue to shape how we deal with money when interacting with each other. Still, little research has explored how these insights are understood by actual digital P2P payments users and can be translated into practical designs to improve current digital P2P payment systems. As our first endeavor to address this gap, the study adopted a Participatory Design approach to investigate how actual users envision and redesign future digital P2P payments to better facilitate their social connections with known contacts. Our study contributes to the current HCI discourse on the increasing interplay of financial technology and social connections by (1) uncovering nuanced user perceptions of digital P2P payments in everyday social interactions and (2) offering creative and actionable future directions to redesign digital P2P payments to further facilitate social connections based on user-generated design envisions.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

KEYWORDS

digital P2P payments, interpersonal relationships, social connections, research through design, participatory design

ACM Reference Format:

Lingyuan Li, Guo Freeman, and Wen Duan. 2024. Exploring Redesigning Digital P2P Payments to Facilitate Social Connections: A Participatory Design Approach. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3651095>

1 INTRODUCTION

Money transactions, including spending, lending, borrowing, and gifting, are integral to interpersonal communication [20]. Therefore, the ways in which people exchange money can significantly affect

their interpersonal relationships [5, 15, 27–29, 51]. More recently, digital P2P payments (e.g., Venmo, MobilePay, PhonePe, WeChat Pay, Apple Pay) have risen to prominence as a preferred method for conducting online money transactions among friends, family, and colleagues [10]. Particularly, the ubiquity and seamless nature of these services in our daily lives and their ability to facilitate direct money transactions through computer-mediated channels seem to pose significant influences on people's interpersonal relationships when money is involved. Indeed, previous research has explored various impacts of digital P2P payments on social relationships. For example, findings from a 2018 Zelle survey indicated that a majority of users would consistently follow up with friends to retrieve shared expenses [50]. Research by Tang et al. suggested that digital transactions via platforms like Venmo and WeChat Pay could ease social awkwardness in friendships [43]. Park et al. highlighted the tendency to use digital apps for debt communication, especially with weaker social ties [34]. Our prior research [27–29] has also expanded on these findings and delved deeper in exploring the perceptions and influences of digital P2P payments in existing offline interpersonal relationships among *known contacts* (i.e., people who already knew each other, such as friends, family members, colleagues, and neighbors).

Taken together, these works have highlighted the importance of further understanding the growing role of digital P2P payments in our networked lives through (re)shaping how we talk about and conduct money transactions when interacting with each other. Still, little research has explored how these insights are understood by actual digital P2P payments users and can be translated into practical designs to improve the current digital P2P payment platforms. As our first endeavor to address this gap, our study aims to leverage the Research through Design (RtD) methodology with an emphasis on Participatory Design (PD) approach to explore the following research question:

RQ: How would actual users envision and redesign future digital P2P payments to better facilitate their social connections with known contacts?

As an initial step in exploring the redesign of digital P2P payments, the study begins the important work of bridging actual user perspectives with the design of digital financial tools. Our study thus contributes to the current HCI discourse on the increasing interplay of financial technology and social connections. First, we offer empirical evidence of how actual users envision the future of digital P2P payment services and uncover nuanced user-generated understandings of how these services are perceived in the context of everyday social interactions. The PD approach we adopted has also yielded a range of user-generated design recommendations,

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3651095>

which offer creative and actionable future directions to redesign digital P2P payments to further facilitate social connections.

2 TECHNOLOGY-SUPPORTED MONEY TRANSACTIONS AND INTERPERSONAL RELATIONSHIPS

The popularity of digital banking, online payments, and virtual currencies has intricately linked technology-supported financial transactions with social interactions, sparking interest in HCI research. This area explores the integration of financial interactions in digital and new media [20], innovative approaches to managing personal finance digitally [26], and the concept of "Financial Citizenship" in contemporary society [3]. Existing research has underscored the complex role of technology-mediated financial transactions in shaping interpersonal relationships, with the potential to both hinder and support their development [31]. Wherry et al. highlighted the awkwardness and strained reciprocity in lending money among friends and relatives [47], while a Zelle survey (2018) indicated the persistence in reclaiming money lent to friends [50]. Jiang et al. found that monetary involvement often leads to more business-like, less emotional interactions [17]. Conversely, financial gestures like donations in live streaming are seen as vital social support [48]; virtual gifting on platforms like live streaming and gaming fosters community building and reciprocity [46, 52]; and digital gifting methods, such as WeChat's red packets, also illustrates new ways to strengthen group bonds in China [49]. As financial technologies evolve, the nuances of how people integrate money transactions in their interpersonal relationships through technology continue to diversify [20].

In particular, **digital P2P payments** involve transferring money **between two individuals** using banking accounts or credit/debit cards through an online or mobile service for everyday activities [5, 44]. Distinct from other digital payment methods such as online payments (which typically involve transactions *between customers and merchants* for utilities, traffic tickets, or online shopping [13, 22, 24, 32]) and mobile payments (focused on *using mobile devices for transactions with both individuals and companies* [8]), digital P2P payments are unique in their focus on interpersonal financial exchanges. The burgeoning use of digital P2P payments, further accelerated by the pandemic, has become a focal point of interest extensively explored in HCI research, particularly the social implications of digital P2P payment apps. For instance, studies have investigated how platforms like Venmo support both financial transactions and social interactions, with unique aspects like payment earmarking, emoji usage, and social activity stream [1, 2, 6]. Other research has probed into factors influencing app usage, such as trust, usefulness, and the role of social relationships [16, 19, 51]. While existing studies highlight the important role of digital P2P payments in transactions and socialization, our research takes a step further. Building upon existing works that mainly focus on analyzing digital P2P payments' impacts on human relationships, this work represents our first endeavor to provide actionable insights for redesigning future digital P2P payment services to further facilitate social connections among known contacts, especially through actively involving actual users in the redesign process.

3 METHODS

3.1 Research through Design (RtD) and Participatory Design (PD) Approach

To address our RQ, we adopted the RtD methodology with an emphasis on PD approach [40, 53] in this work. We consider RtD appropriate and necessary for answering our RQ because it is a well-recognized HCI methodology for integrating empirical research and design to leverage insights from design practice to tackle complex, future-oriented design challenges [14, 54]. PD, a key component of RtD, involves users directly in the design process, fostering collaboration and co-creation [38]. In this sense, the use of PD aims to reconfigure the relationships between technology designers and users and align technology with how people desire to conduct their activities [7, 21]. As we are especially motivated to ensure the central role of actual users' perspectives in the (re)design process of digital P2P payment services, we believe that this approach aligns well with our key research goal.

3.2 Co-designers

Co-designers Recruitment. This study is part of a broader multi-year project examining the impact of digital P2P payments on interpersonal relationships. We invited participants from the larger project to join this particular PD study on redesigning digital P2P payments to support social connections between known contacts. A snowball sampling was also used to recruit participants. Five individuals agreed to participate as "co-designers," a term used to describe non-professionals who collaborate in the design process [39]. All co-designers self-identified as experienced users of digital P2P payment services. Their in-depth understanding and insider knowledge of digital P2P payments thus could bring in valuable insights and profound personal experiences to the redesign process.

Co-designers Demographics. Out of the 5 co-designers, 3 self-reported as women and 2 as men. They were aged 24 to 30 years old with an average age of 26.5. 3 self-reported as Asian (2 as Chinese, 1 as Indian) and 2 as White. Table 1 summarizes the demographic information of co-designers, along with information about digital P2P payment services they had used, how long they had used such services, and the places where they grew up and live now.

3.3 Participatory Design Activities

Our review of the existing literature on social aspects of digital P2P payments [2, 16, 19, 43, 45] combined with insights from our prior research [27–29], reveals critical aspects of how using digital P2P payments impacts social connections with known contacts. Specifically, these impacts include: 1) *reduced awkwardness* in transactions with known contacts; 2) *enhanced sense of fairness* in group payment management; 3) *increased trust* in known contacts; 4) *decreased tensions* when talking about money; 5) *loss of emotion* in communication; 6) *increased peer pressure* to adopt specific payment platforms; and 7) *potential decrease in emotional attachment* with known contacts. Informed by these insights, we designed and conducted two rounds of PD activities to specifically explore how actual users would envision and redesign future digital P2P payments to amplify the identified positive effects (e.g., reducing

ID	Gender	Age	Self-Reported Ethnicity	Digital P2P Payments Used	Experiences (Years)	Place where Grew Up	Area of Residency
C1	Woman	27	White	Venmo, Apple Pay	7	Suburban	Between suburban and rural
C2	Man	24	Asian	Venmo	6	Suburban	Suburban
C3	Woman	N/A	Asian	WeChat Pay, Alipay, Zelle, Venmo, PayPal	10	Urban	Suburban
C4	Woman	30	Asian	WeChat Pay, Alipay, Zelle, Venmo, Cash App	10	Urban	Suburban
C5	Man	25	White	Venmo	12	Rural	Between suburban and rural

Table 1: Demographic information of co-designers. (Note: N/A – participants preferred not to answer.)

awkwardness) while mitigating negative impacts (e.g., enhancing emotional communication).

3.3.1 Round 1. We conducted the first round of PD activities in a hybrid way in December 2022. Four of the five co-designers attended in person and one joined via Zoom. Round 1 PD activities lasted 2 hours and 8 minutes in total and involved two essential phases: warm-up; and brainstorming and ideation.

Warm-up. Invited to discuss as a group, 5 co-designers were briefed on the study overview, consent document, and the overall structure and process of the PD activities. After receiving their verbal consent, we initiated an audio recording of the session. Co-designers freely shared their personal experiences of using digital P2P payments and reflected upon any impacts of digital P2P payments on social relationships between known contacts. This discussion thus well prepared the co-designers for the creative process and grounded their following ideation in real-world experiences.

Brainstorming and Ideation. We utilized *Crazy 8's*, a design sprint method that promotes rapid and divergent idea generation [12]. Specifically, we framed ‘How Might We’ questions of Crazy 8’s based on the insights from prior research on digital P2P payments’ impacts on people’s social connections to guide co-designers in the brainstorming process (e.g., “How might we reduce peer pressure during transactions with known contacts via digital P2P payments?”). By doing so, we aimed to ensure that the co-designers’ envisions aligned with the overarching goal of supporting social connections through redesigning digital P2P payments. Co-designers individually sketched design ideas and collectively discussed and identified the most promising and valuable designs to them (see Figure 1).

3.3.2 Creating Initial Prototypes. Based on co-designers’ collaborative decision-making on redesigning digital P2P payments to better support social connections in their own views, we then created the clickable, interactive prototypes of the most compelling and valuable design ideas they voted on by using Figma (a collaborative interface design tool) [42]. This step transformed co-designers’ high-level concepts into more tangible designs, allowing for more concrete reflections and discussions in Round 2.

3.3.3 Round 2. In January 2023, the second round of PD activities was conducted online with the same participants via Zoom. This round lasted 47 minutes, with both video and audio recorded. It commenced with a recap and reflection on Round 1 of PD activities. We then presented the clickable, interactive prototypes for co-designers to collaboratively review and reflect on their design ideas in a more tangible way. The co-designers offered insights on how well these designs met their envisions. They then proposed adjustments, additional refinements, and new elements to enhance the

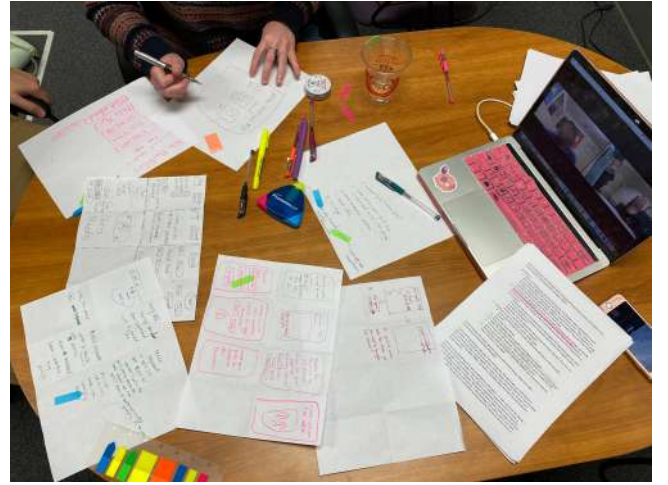


Figure 1: Co-designers sketched their design ideas individually and voted their favourite ones.

designs. Importantly, the objective of this round was not to achieve finalized, complete prototypes but to iteratively refine and expand upon the initial design visions. In this sense, our approach highlights the iterative nature of design thinking inherent in RtD, where insights are not merely extracted from discussions but also through hands-on interaction with design prototypes. This methodological choice underscores the added value of RtD and PD beyond traditional interview-based research on digital P2P payments, which provides a rich, nuanced understanding of user needs, preferences, and the contextual applicability of design solutions.

3.4 Data Analysis

Upon completing both rounds of PD activities, recordings were transcribed for further data analysis. We used an empirical, in-depth qualitative thematic analysis [4] of the transcribed data to identify user needs and summarize design solutions. In doing so, we first comprehensively reviewed the transcriptions to grasp co-designers’ expectations and proposed ideas. We then assigned codes to pertinent data, streamlined redundancies, and merged similar codes into thematic categories under (1) underlying user needs for using digital P2P payments for social connections and (2) design ideas to support such needs. Through a collaborative, iterative coding process, we refined these themes and generated a coherent narrative regarding participants’ envisions and design suggestions.

4 RESULTS

Through two rounds of PD activities, we identified 29 design ideas envisioned by the co-designers to redesign future digital P2P payments to facilitate people's social connections with known contacts. We also pinpointed 10 underlying user needs driving these ideas. These user needs center four key envisions: reducing awkwardness, enhancing fairness, decreasing peer pressure, and increasing emotion.

4.1 Envision 1: Understanding and Designing for Reducing Awkwardness

Our PD activities revealed five key user needs aiming at minimizing perceived awkwardness in digital P2P payment transactions with known contacts. We also summarized the co-designers' proposed design ideas for these needs.

Need 1. Avoiding disputes or ambiguity in shared activities and bills: The co-designers identified the importance of clarity and mutual understanding in transactions of shared activities and bills, as confusion and disputes often stem from a lack of awareness about shared financial engagements. A proposed solution, as illustrated by C3, involves a feature that records and displays shared expenses, like *"if I go to have sushi with [a friend] and I can add her, then she sees it on her 'My Spend Today.'"* Such a feature would not only facilitate mutual awareness but also subtly prompt repayment, reducing potential ambiguity and conflicts. In Round 2, an enhanced interface design for easy access and review of these records was also suggested to minimize misunderstandings and maintain positive relationship dynamics.

Need 2. Having diplomatic reminders in shared transactions: The co-designers emphasized the importance of tactful reminders sent by digital P2P payment platforms for owed money, which would help address the discomfort of personally reminding known contacts about shared expenses. They advocated for discreet, system-generated notifications to reduce the personal aspect of such reminders. C3 captured this sentiment: *"a system notification like 'Hey, you had sushi with [name] today. Do you need to transfer her money?'"*, offering a less confrontational way to remind about repayments. C1 proposed customizable urgency levels for these reminders, sent as official platform messages rather than direct personal prompts, allowing users to tactfully nudge others about repayments without causing embarrassment (see Figure 2). Furthermore, automated notifications related to shared activity records could also act as subtle reminders, helping to alleviate the awkwardness associated with direct monetary reminders in personal relationships. This need highlights the value of an impersonal yet effective reminder mechanism in digital P2P payment platforms.

Need 3. Introducing fun elements in transactions: The co-designers highlighted that finance-related activities, particularly between known contacts, often carry an inherent awkwardness due to their sensitive and judgmental nature. There was a consensus on the need to make financial interactions more captivating and informal, thereby diminishing the awkwardness by infusing a sense of fun into the transaction process. To achieve this, C4 suggested (see Figure 3), *"you have this virtual physical bill. You can make that into different shapes [...] So the funniness will reduce the awkwardness."* Likewise, C2's idea of customizing transaction alerts with unique

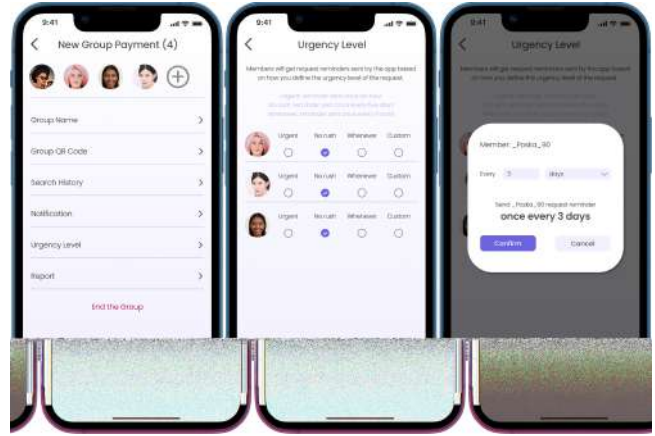


Figure 2: People can customize different urgency levels of reminders. (Details visible upon zooming)

ringtones added to this light-hearted approach. These design solutions were also positively received in Round 2 to further confirm that such playful elements were effective in making digital P2P payment transactions more enjoyable and less awkward between known contacts.

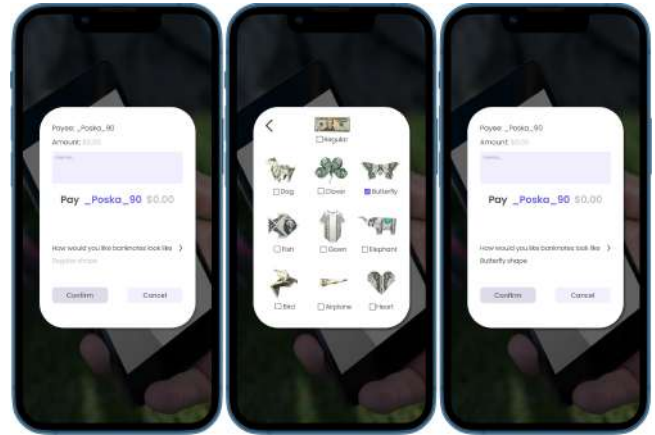


Figure 3: People can choose which shape they expect the virtual banknotes to look like. (Details visible upon zooming)

Need 4. Promoting healthy-manner transactions: The co-designers stressed the need to promote healthy financial behaviors (e.g., repaying in a timely manner, without reminders or even proactively) in digital P2P transactions among known contacts. Co-designers sought to foster a culture of punctuality and responsibility, reducing awkwardness and potential conflicts. One suggested method to encourage this behavior is the implementation of reward mechanisms for prompt responses to payment requests. C2 proposed *"rewards associated with spending money,"* where users who promptly complete transactions could receive incentives. Additionally, C5 linked this to a trust level system inspired by social media metrics, like Snapstreaks: *"If I request money from you, like an 87 Snapstreak with you, you become a trusted individual, with bragging*

rights for timely payments." In Round 2, co-designers expanded on this idea by suggesting a trust score system for digital P2P platforms, which is similar to the FICO score - a credit score used by U.S. financial institutions to evaluate an individual's credit risk and determine how likely they are to repay a loan on time. This system would feature an app-wide trust ranking to demonstrate users' reliability in making money transactions that they should make. In doing so, co-designers hoped to both minimize discomfort in personal transactions and motivate users to uphold their financial reputation within their social circles (see Figure 4).



Figure 4: People can check their friends' trust scores and the trust score rank. (Details visible upon zooming)

Need 5. Enhancing bill splitting nuances: Reflecting on our previous research [27–29] on the awkwardness and discomfort of discussing bill splitting among known contacts, the co-designers agreed on the need for more subtle and flexible ways in digital P2P payments to split bills. They suggested that doing so could help minimize the potential embarrassment associated with dividing expenses. C5 thus proposed the concept of group payment "Bar Tab": *"When you pay for things, your bar tab pays for it, and then it naturally divvies out transactions to the individuals who are prescribed to the bar tab."* This concept features straightforward, one-click options for splitting bills based on different scenarios and preferences, such as evenly dividing shared costs or allowing group members to define their shares to pay. Additionally, the co-designers discussed incorporating a group chat function for real-time communication in Round 2. This feature would facilitate further discussion and clarification among group members regarding payment details in bill splitting.

4.2 Envision 2: Understanding and Designing for Ensuring Fairness

We identified one user need for ensuring fairness in digital P2P payments with known contacts. We also summarized the co-designers' proposed design ideas for the need.

Need 6. Simplifying the dynamics and easing responsibilities in group payments: A key user need emerges from solutions for fairness in digital P2P payments: reducing the burden on individuals typically responsible for group payments and subsequent

reimbursements. This is essential for fostering fair and balanced relationships in digital P2P transactions. The co-designers proposed two primary solutions. The first is *selective payment*. Users should only pay for what they have spent. For instance, a feature where a scanned receipt allows the system to detect items, enabling users to easily choose and pay for only their items (see Figure 5). This thus simplifies the process and ensures everyone pays their share fairly. The second is *pre-planned group payments*. Instead of one person paying upfront, C5 suggested a "Plan Ahead" feature (see Figure 6). This involves group members depositing money in a collective account in anticipation of an event's cost. The idea, refined in Round 2 discussions, envisions that the account can be adjusted post-event to match the actual spending, with members receiving refunds or making additional contributions as necessary. These ideas aim to streamline group payments, ensuring fairness and easing the burden on any one person in digital P2P payment scenarios.

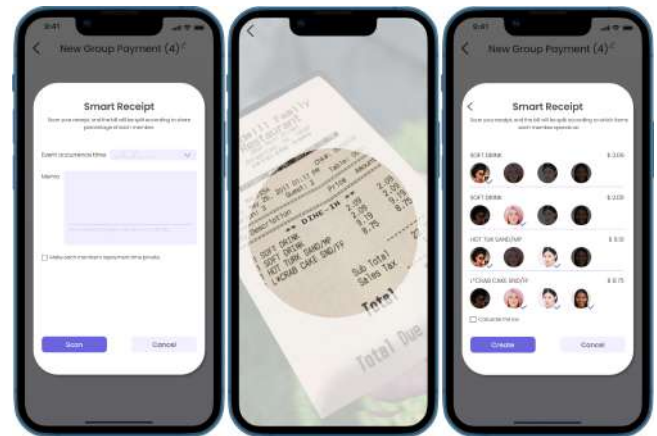


Figure 5: People can easily split bills by scanning their receipt and only select items they need to pay. (Details visible upon zooming)

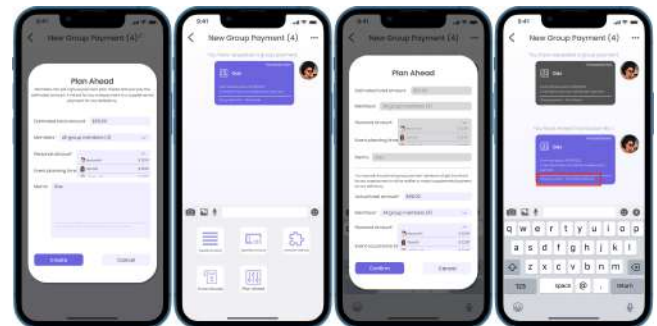


Figure 6: People can set up "Plan Ahead" and revisit it to edit the amount they actually spent. (Details visible upon zooming)

4.3 Envision 3: Reducing Peer Pressure in Digital P2P Payments Adoption

We identified two user needs regarding decreasing peer pressure in digital P2P payment transactions. In particular, the needs mainly focus on how to make people more willing to use digital P2P payments rather than being forced to use them due to peer pressure.

Need 7. Enhancing appeal and willingness to use digital P2P payments: To reduce peer pressure in adopting digital P2P payments, the co-designers identified the need to enhance their appeal and willingness to use. One of the reasons for reluctance to use digital P2P payments often stems from the unwillingness to download the apps, especially for people who are unfamiliar with them [29]. Co-designers suggested making digital P2P payment services more attractive by offering secondary benefits. C2 pointed out, *"there's a secondary benefit not only to this transaction, but you're actually doing something that helps the world."* Incentives like earning points from transactions, contributing to environmental causes, or enabling donations through the app could motivate users. In Round 2, the co-designers elaborated on this idea and proposed that users could use these points for rewards or to offset transaction costs (see Figure 7), thereby shifting perceptions from obligation to personal benefit and interest.

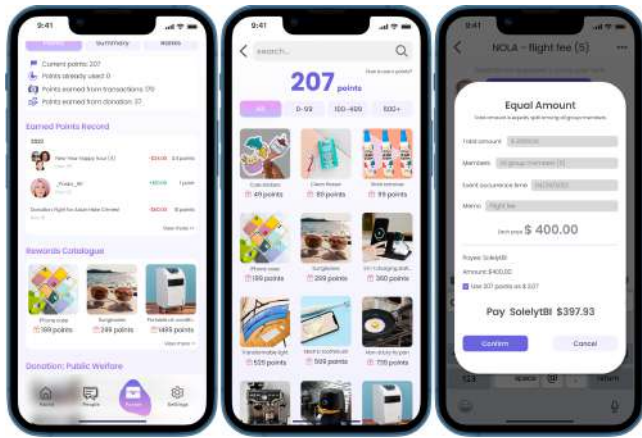


Figure 7: People can use points they earned to redeem rewards or offset part of the amount they need to pay in money transactions. (Details visible upon zooming)

Need 8. Prioritizing privacy in digital P2P payment services: Prior research, including our own [27, 29], has highlighted that addressing privacy concerns is crucial for increasing the adoption of digital P2P payments, especially among non-smartphone users and older adults wary of online transactions. The critical issue is a lack of trust in how the involved personal information can be handled in a secure way. To mitigate these concerns, digital P2P payments must prioritize user privacy. C1 recommended, *"I think when you sign up for one of these apps the very first time, you should basically pick privacy settings where you can."* This idea suggests implementing customizable privacy settings during the initial app setup, offering users greater control and reassurance about their personal data security. For the co-designers, this focus on privacy

is essential to build trust and encourage the broader use of digital P2P payment platforms.

4.4 Envision 4: Fostering Emotional Connectivity in Digital P2P Payments

Two underlying user needs emerged regarding increasing emotion in communication in digital P2P payments with known contacts. We also summarized the co-designers' proposed design ideas for these needs.

Need 9. Adding emotional depth and personalization: The co-designers echoed with previous findings, highlighting the tendency of digital P2P payments to diminish the human element in money transactions. They then identified the need to infuse emotional depth and personalization into digital P2P payments. For instance, C5 suggested that emoticons (smiley, medium, and sad faces) should be incorporated into digital P2P payments to reflect users' feelings about each transaction. To counter the loss of contextual cues and emotional expressions, co-designers proposed features to personalize emotional expressions within each transaction, such as: *"send nuanced thank-yous, especially in gift giving"* (C1). Expanding this, others also suggested using various expressive formats in messages, like video, audio, text, or images, for more vivid and explicit communication in digital transactions (see Figure 8).

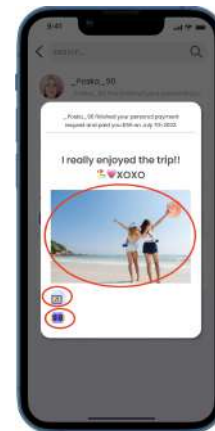


Figure 8: People can attach pictures and video/audio clips in the memo when they make personal money transactions. (Details visible upon zooming)

Need 10. Simulating the physicality of face-to-face cash transactions: The co-designers valued the tangible aspect of cash transactions. Consequently, replicating the sense of physicality present in these traditional transactions is vital to better embody the emotional nuances of monetary exchanges. To realize this, several designs were proposed to mirror offline cash transaction habits. C4's idea involves the use of physicalized virtual representations of currency, for example, selecting ten \$5 bills virtually instead of simply typing in "\$50." Another concept is to simulate the act of handing over cash through augmented reality (AR) technology (see Figure 9). Further enhancing this sensory experience, C4 proposed: *"After you pay for the meal, you tap your phone, it comes with*

an interesting vibration, like a hug," aiming to add a personal "touch" by generating a device vibration akin to a hug. In Round 2, the co-designers were optimistic about these designs, believing that integrating such features could heighten users' awareness of the physicality of transactions and foster a deeper personal connection between the sender and recipient through digital P2P payments.



Figure 9: Using AR to simulate the sense of physicality of face-to-face cash transactions via digital P2P payments. (Details visible upon zooming)

5 DISCUSSION AND FUTURE WORK

Recognizing the critical role of digital P2P payment systems in modern social interactions, our study has unveiled essential user needs and envisions for the future of digital P2P payments to better facilitate their social lives. Although previous studies have touched upon digital P2P payments users' online social patterns [1, 2, 6, 45, 51], our findings specifically highlight how effectively and uniquely the envisioned digital P2P payments can enable users to manage money transactions while facilitating offline interpersonal relationships with known contacts.

First, our research extends existing understanding by showing how digital P2P payments can alleviate the sensitivity and embarrassment often associated with cash transactions [9, 11, 30]. Our findings suggest actionable design directions to further reduce awkwardness through improved clarity, diplomatic communication, engaging elements, promoting financial responsibility, and flexible bill splitting. This may inform the design of future digital P2P payments to facilitate more socially sensitive and enjoyable financial transaction experiences, which will enhance both usability and interpersonal dynamics with known contacts. Second, our participants' envisions also highlight the need to create more equitable and stress-free group payment experiences to ensure fairness in shared financial activities. This can be achieved by simplifying group payment dynamics and distributing financial responsibilities more evenly, which fosters a sense of fairness and collective accountability. Such a sense is especially crucial for maintaining balanced and harmonious relationships with known contacts. Third, echoing challenges in adopting digital financial technologies [18, 33], our participants further highlight the need to transform digital P2P

payments into a choice driven by personal preference and trust [19], rather than pressure from their social circles. By increasing the appeal and willingness to use digital P2P payments and prioritizing privacy, such a shift has the potential to foster a more inclusive and trustworthy digital payment environment. Fourth, while previous studies have acknowledged the transformative nature of computer-mediated money transactions [25, 36, 37], our study has especially identified novel ways to infuse money transactions with a sense of personal touch and emotional significance. By adding emotional depth and simulating the physicality of face-to-face cash transactions, our participants envision that future digital P2P payments can go beyond just offering financial exchanges but provide richer and more meaningful offline interpersonal interactions with known contacts through such transactions.

Furthermore, the scalability of such design ideas across diverse cultural landscapes remains an open question. For instance, the acceptability of splitting bills or lending money among known contacts varies significantly across cultures. In Iceland and Norway, splitting bills is considered polite, reflecting a preference for financial independence and a desire not to feel indebted to anyone [35]. Conversely, in Vietnam, Lebanon, Iran, Mexico, and China, such practices might signal a breach of social etiquette, where covering the entire bill is often seen as a gesture of goodwill or friendship [23, 41]. This discrepancy poses challenges in developing digital P2P payment solutions that are adaptable to various cultural norms and social expectations worldwide. Moving forward, we aim to broaden our study to include a more diverse participant pool across different demographics (e.g., ages, countries, and ethnicities) to deepen our exploration of culturally sensitive design. We also plan to conduct a systematic evaluation of our prototypes, including both quantitative and qualitative considerations to further enhance the redesign of future digital P2P payments. In an era where digital technologies are profoundly reshaping social interactions, we hope that our work will contribute to further understanding and designing for the role of digital P2P payments in facilitating not just financial transactions but also everyday interpersonal relationships.

ACKNOWLEDGMENTS

We thank our participants for their time and input and the anonymous reviewers for their useful suggestions. We especially thank Donghee Yvette Wohn, Bart Knijnenburg, and Nathan J. McNeese for their valuable insights that help structure our research design.

REFERENCES

- [1] Amelia Acker and Dhiraj Murthy. 2018. Venmo: Understanding mobile payments as social media. In *Proceedings of the 9th international conference on social media and society*. 5–12.
- [2] Amelia Acker and Dhiraj Murthy. 2020. What is Venmo? A descriptive analysis of social features in the mobile payment platform. *Telematics and Informatics* 52 (2020), 101429.
- [3] Belén Barros Barros Pena, Bailey Kursar, Rachel E Clarke, Katie Alpin, Merlyn Holkar, and John Vines. 2021. Financial Technologies in the Cycle of Poor Mental Health and Financial Hardship: Towards Financial Citizenship. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [4] Virginia Braun and Victoria Clarke. 2012. *Thematic analysis*. American Psychological Association.
- [5] Andrea J Caceres-Santamaria. 2020. Peer-to-Peer (P2P) Payment Services. *Page One Economics*® (2020).
- [6] Monica Caraway, Daniel A Epstein, and Sean A Munson. 2017. Friends don't need receipts: The curious case of social awareness streams in the mobile payment app Venmo. *Proceedings of the ACM on Human-Computer Interaction* 1, CSCW (2017), 1–17.

- [7] John M Carroll. 1996. Encountering others: Reciprocal openings in participatory design and user-centered design. *Human-computer interaction* 11, 3 (1996), 285–290.
- [8] Tomi Dahlberg, Niina Mallat, Jan Ondrus, and Agnieszka Zmijewska. 2008. Past, present and future of mobile payments research: A literature review. *Electronic commerce research and applications* 7, 2 (2008), 165–181.
- [9] Gwarlann De Kerviler, Nathalie TM Demoulin, and Pietro Zidda. 2016. Adoption of in-store mobile payment: Are perceived risk and convenience the only drivers? *Journal of Retailing and Consumer Services* 31 (2016), 334–344.
- [10] Donna Fuscaldo. 2019. Here's An Area Of Digital Payments That's Booming: P2P Payments. *Forbes*. August (2019).
- [11] Erika Giovanetti. 2020. 31% of Americans Would Rather Go Into Debt Than Borrow From Loved Ones. <https://www.lendingtree.com/personal/lending-between-friends-and-family-survey>.
- [12] Voula Gkatzidou, Joseph Giacomini, and Lee Skrypchuk. 2021. Automotive Human Centred Design Methods. In *Automotive Human Centred Design Methods*. De Gruyter.
- [13] GoCardless. 2023. Online Payments for Business - An introduction to online payments. <https://gocardless.com/guides/online-payments-guide/online-payments-introduction/>.
- [14] Danny Godin and Mithra Zahedi. 2014. Aspects of research through design: a literature review. (2014).
- [15] Lou Grilli. 2017. P2P – A comprehensive look at person-to-person payments. <http://www.thepaymentsreview.com/a-look-at-p2p-payments>.
- [16] Lucas Heijting. 2020. Just send me a payment request: what factors determine the behavioural intention to use P2P payment applications? (2020).
- [17] Yuwei Jiang, Zhansheng Chen, and Robert S Wyer Jr. 2014. Impact of money on emotional expression. *Journal of Experimental Social Psychology* 55 (2014), 228–233.
- [18] K Kajol, Ranjit Singh, and Justin Paul. 2022. Adoption of digital financial transactions: A review of literature and future research agenda. *Technological Forecasting and Social Change* 184 (2022), 121991.
- [19] Zoran Kalinic, Veljko Marinkovic, Sebastian Molinillo, and Francisco Liebanacabanillas. 2019. A multi-analytical approach to peer-to-peer mobile payment acceptance prediction. *Journal of Retailing and Consumer Services* 49 (2019), 143–153.
- [20] Joseph Jofish Kaye, Mary McCuiston, Rebecca Gulotta, and David A Shamma. 2014. Money talks: tracking personal finances. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 521–530.
- [21] Finn Kensing and Jeanette Blomberg. 1998. Participatory design: Issues and concerns. *Computer supported cooperative work* 7, 3-4 (1998), 167–185.
- [22] Burhan UI Islam Khan, Rashidah F Olanrewaju, Asifa Mehrab Baba, Adil Ahmad Langoo, and Shahul Assad. 2017. A compendious study of online payment systems: Past developments, present impact, and future considerations. *International journal of advanced computer science and applications* 8, 5 (2017).
- [23] Yong Ming Kow, Xinning Gui, and Waikuen Cheng. 2017. Special digital monies: The design of alipay and wechat wallet for mobile payment practices in china. In *Human-Computer Interaction-INTERACT 2017: 16th IFIP TC 13 International Conference, Mumbai, India, September 25-29, 2017, Proceedings, Part IV* 16. Springer, 136–155.
- [24] Pine Labs. 2022. What is Online Payment? Types, Modes, Methods, Meaning. <https://www.pinelabs.com/blog/online-payments-and-its-types-methods-and-meaning>.
- [25] Lucas Lengacher. 2015. Mobile technology: Its effect on face-to-face communication and interpersonal interaction. *Undergraduate Research Journal for the Human Sciences* 14, 1 (2015).
- [26] Makayla Lewis and Mark Perry. 2019. Follow the money: Managing personal finance digitally. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–14.
- [27] Lingyuan Li and Guo Freeman. 2021. Money vs. Social Life: Why People Choose Not to Use Facebook Messenger Payment. (2021).
- [28] Lingyuan Li, Guo Freeman, and Bart Knijnenburg. 2024. Beyond Just Money Transactions: How Digital P2P Payments (Re)shape Existing Offline Interpersonal Relationships. *Proceedings of the ACM on Human-Computer Interaction* 8, CSCW1 (2024), 1–36.
- [29] Lingyuan Li, Guo Freeman, and Donghee Yvette Wohn. 2021. The interplay of financial exchanges and offline interpersonal relationships through digital peer-to-peer payments. *Telematics and Informatics* 63 (2021), 101671.
- [30] Niina Mallat. 2007. Exploring consumer adoption of mobile payments—A qualitative study. *The Journal of Strategic Information Systems* 16, 4 (2007), 413–432.
- [31] Nicole L Mead and Anika Stuppy. 2014. Two sides of the same coin: Money can promote and hinder interpersonal processes. In *The psychological science of money*. Springer, 243–262.
- [32] Tatiana Morand. 2023. The Best Online Payment Service Providers in 2023. <https://www.wildapricot.com/blog/online-payment-services>.
- [33] Catarina Neves, Tiago Oliveira, Fernando Santini, and Luis Gutman. 2023. Adoption and use of digital financial services: A meta analysis of barriers and facilitators. *International Journal of Information Management Data Insights* 3, 2 (2023), 100201.
- [34] Alexander Park, Cynthia Cryder, and Rachel Gershon. 2020. Fighting Fiscal Awkwardness: the (Dis) Advantages of Digital Payment Methods on Peer-Debt Dynamics. *ACR North American Advances* (2020).
- [35] Sean Percival. 2020. *Working with Norwegians*. Vol. 2. Stordale OU.
- [36] Ronald E Rice and Gail Love. 1987. Electronic emotion: Socioemotional content in a computer-mediated communication network. *Communication research* 14, 1 (1987), 85–108.
- [37] Everett M Rogers et al. 1986. *Communication technology*. Simon and Schuster.
- [38] Elizabeth B-N Sanders. 2002. From user-centered to participatory design approaches. In *Design and the social sciences*. CRC Press, 18–25.
- [39] Elizabeth B-N Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *Co-design* 4, 1 (2008), 5–18.
- [40] Douglas Schuler and Aki Namioka. 1993. *Participatory design: Principles and practices*. CRC Press.
- [41] Elisabeth Sherman. 2022. Dining Etiquette: How To Split the Bill in 26 Countries Around the World. <https://matadornetwork.com/read/splitting-the-bill-etiquette-around-the-world>.
- [42] Fabio Staiano. 2022. *Designing and Prototyping Interfaces with Figma: Learn essential UX/UI design principles by creating interactive prototypes for mobile, tablet, and desktop*. Packt Publishing Ltd.
- [43] Shiliang Tang, Ziming Wu, Xinyi Zhang, Gang Wang, Xiaojuan Ma, Haitao Zheng, and Ben Zhao. 2019. Towards understanding the adoption and social experience of digital wallet systems. In *Proceedings of the 52nd Hawaii International Conference on System Sciences*.
- [44] Dillon Thompson. 2019. Banking 101: What Are Peer-to-Peer (P2P) Payments? <https://www.depositaccounts.com/blog/p2p-payments.html>.
- [45] Clive Unger, Dhiraj Murthy, Amelia Acker, Ishank Arora, and Andy Chang. 2020. Examining the evolution of mobile social payments in Venmo. In *International Conference on Social Media and Society*. 101–110.
- [46] Yang Wang and Scott D Mainwaring. 2008. Human-Currency Interaction: learning from virtual currency use in China. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 25–28.
- [47] Frederick F Wherry, Kristin S Seefeldt, and Anthony S Alvarez. 2019. To lend or not to lend to friends and kin: Awkwardness, obfuscation, and negative reciprocity. *Social Forces* 98, 2 (2019), 753–793.
- [48] Donghee Yvette Wohn, Guo Freeman, and Caitlin McLaughlin. 2018. Explaining viewers' emotional, instrumental, and financial support provision for live streamers. In *Proceedings of the 2018 CHI conference on human factors in computing systems*. 1–13.
- [49] Ziming Wu and Xiaojuan Ma. 2017. Money as a social currency to manage group dynamics: Red packet gifting in chinese online communities. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. 2240–2247.
- [50] Zelle. 2018. Early Warning Holiday Data. <https://www.zellepay.com/sites/default/files/2019-04/Early%20Warning%20Holiday%20Data%20PPT%20v3.pdf>.
- [51] Xinyi Zhang, Shiliang Tang, Yun Zhao, Gang Wang, Haitao Zheng, and Ben Y Zhao. 2017. Cold hard e-cash: Friends and vendors in the venmo digital payments system. In *Eleventh International AAAI Conference on Web and Social Media*.
- [52] Xiaoxing Zhang, Yu Xiang, and Lei Hao. 2019. Virtual gifting on China's live streaming platforms: Hijacking the online gift economy. *Chinese Journal of Communication* 12, 3 (2019), 340–355.
- [53] John Zimmerman and Jodi Forlizzi. 2014. Research through design in HCI. *Ways of Knowing in HCI* (2014), 167–189.
- [54] John Zimmerman, Jodi Forlizzi, and Shelley Evenson. 2007. Research through design as a method for interaction design research in HCI. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 493–502.